Release Notes

CY7C652xx USB-Serial Software Development Kit for Windows

Release Date: September 30, 2015

Thank you for your interest in the CY7C652xx USB-Serial LP Software Development Kit (SDK). This document lists installation requirements, software and hardware updates, and other information related to the SDK for Windows.

System Requirements and Recommendations

Hardware/Operating System Requirements	**Minimum**	**Recommended**
Processor speed	1 GHz or faster	2 GHz Dual Core
RAM	512 MB	1 GB or higher
Free hard drive space	1 GB	
Screen resolution	1024×768	
CD/DVD drive	✓	✓
USB	Full-Speed	
Windows OS	Windows XP	Windows XP and above
Software Prerequisites	**Minimum**	**Recommended**
Microsoft .NET Framework	3.5 SP1	
Microsoft VC++ Redistributable	VC++ 2008 SP1	
Adobe Reader (for PDF documents)	9.2*	

* For Windows 7, the minimum required version of Adobe Reader is version 9.2. You can download the latest version here: http://get.adobe.com/reader/

Installation

Download and install the CY7C652xx SDK from http://www.cypress.com/go/usbserial.

Updates

This version of 2.0.3 release of the USB-Serial SDK also includes:

- Support for USB CDC class to SPI / I2C bridging functionality is available in the device and in the windows application library.
- Configuration utility has support to configure and program CDC 2.0.3 silicon parts.
- Windows CDC driver has a special option to hold-on to the same COM port number.
- Windows 10 driver and library support is added. Driver guide documentations are updated to support Windows 10. Windows 10 driver support is available for Virtual COM Port (VCP) driver and USB Vendor Driver.
- Special VCP driver API is added to send user defined time-out for UART BREAK signal.
- Issues related to UART flow control mechanism and UART purge command are fixed in the VCP driver.

SDK version of 1.0.3 release of the USB-Serial SDK also includes:

- CDC Driver and Driver Guide Documentations are updated to support Windows 2000 and Windows 8.1.
- Fixes are applied to the Cypress Library to handle Vendor UART short packet read functionality.
- Memory & Resource leaks are fixed in the USB-Serial Library that are related to Vendor UART / SPI / I2C functionality.

SDK version of the 1.0.1 release of the USB-Serial SDK also includes:

- Device configuration Support is available for the new silicon CY7C64213-28PVXI.
- API, Drivers, configuration utility and Documentation are updated to support this new part release.
- Fixed few issues in the configuration utility that are related to storing the string descriptors.

USB-Serial 1.0 SDK release SDK includes:

- APIs for device initialization, configuration of the serial interfaces, and for data transfer with the serial interface.
- CDC-class driver for UART and Vendor-class driver for SPI, I2C, JTAG, GPIO, and configuration.
- Configuration utility for configuring the various device parameters.
- Test Utility program for testing the SPI/I2C EEPROM flash on the development kit (DVK).
- Documentation – API documentation and Utility user guide.
- Cypress Software License Agreement.

Device Features

- Supports USB-Serial full-speed operation, tested with Intel, Renesas, and Fresco USB 3.0 host controllers.
- Supports configuration of the available serial interface as UART, SPI master, SPI slave, I2C master, I2C slave, and JTAG master. Enables data transfer from USB to external serial peripherals.
- Supports configuration of battery charger detection (BCD).
- Supports configuration of CapSense I/Os.
- Supports configuration of I/Os for functions such as TX LED# and RX LED#.
- Supports configuration of the available GPIOs and uses them as either input or output.

Refer to the device datasheet for more information.

Configuration Utility Features

- Supports configuration of USB vendor ID (VID), product ID (PID), string descriptors, and current requirements.

- Supports configuration of the serial interface to UART, SPI, or I2C.
- Supports configuration of various serial interface parameters such as interface frequency and data width.
- Supports CapSense I/O configuration.
- Supports GPIO configuration.

Refer to the Configuration Utility User Guide for more information.

Known Issues and Solutions

1. Using the same serial number for multiple devices connected to the same PC can result in only one device recognized by the system. So, multiple devices with the same USB VID and PID connected to the same PC should have “No USB Serial Number” or Unique USB Serial Number.
2. Changing the USB interface type (CDC or Vendor) bridging the individual serial control block (SCB) changes the characteristics of the device. Therefore, users are expected to assign a different PID when changing this USB configuration. For example, changing device configuration from USB CDC-class to USB vendor-class needs a different window device driver. So, customers are expected to use different PID for USB CDC configuration verses USB Vendor configuration. Configuration Utility automatically assigns several PID’s based on the device USB configuration. Refer to the Configuration Utility User guide for more information.
3. SPI slave device needs consistent USB bandwidth to transmit data at real time. During our testing, it is observed that SPI slave device operated beyond 1Mbps suffered USB bandwidth starvation. Most PC’s irrespective of the configuration handled SPI slave device up to 1 Mbps of operation.

Silicon Errata

To access the latest versions of the silicon errata, visit http://www.cypress.com/go/usb and navigate to Errata.

Technical Support

For assistance, go to http://www.cypress.com/go/support or contact our customer support at +1(800) 541-4736 Ext. 8 (in the USA), or +1 (408) 943-2600 Ext. 8 (International).

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